

Personalized Static Hand-Gesture Recognition

Classical Machine-Learning Models Compared with Geometric Rules

Gesture Presenter — Machine Learning Course Study

13 July 2026

Abstract

This report presents a personalized, single-participant pilot study built on Gesture Presenter, a desktop presentation controller based on MediaPipe hand landmarks. The study asks whether a lightweight and explainable classical machine-learning model can classify the participant's static gestures more accurately than the application's manually designed geometric rules. Camera frames were processed locally and were not saved; only 21 three-dimensional hand landmarks and experimental metadata were recorded. A class-balanced decision tree, random forest, and logistic regression model were trained on two recording sessions and evaluated on a held-out third session. The random forest performed best, reaching 98.8% frame-level and 98.5% trial-level accuracy. Logistic regression achieved 95.7% and 94.8%, the single tree achieved 91.7% and 93.3%, and the hard-coded rules achieved 80.0% and 79.3%. The largest practical difference was rejection of non-command movement: every learned model performed substantially better than the rules, and the tree recognized every held-out `none` trial, whereas the rules recognized none of them. The result supports the feasibility of personalized learned gesture boundaries, but it does not establish generalization to other users or action-level superiority in the live product.

1 System context

Gesture Presenter converts hand movements into presentation actions such as pointer movement, clicks, scrolling, slide changes, and media control. Its recognition pipeline has two distinct parts:

1. MediaPipe Hand Landmarker detects a hand and returns 21 landmarks, each with normalized x , y , and z coordinates.
2. Application-specific logic maps those landmarks to gestures and then uses temporal state machines to dispatch actions.

The production gesture classifier is not a learned classifier. It applies manually chosen conditions to joint angles, normalized fingertip distances, finger-extension states, thumb direction, and finger gaps. For example, an index pinch is identified through the normalized distance between the thumb and index fingertips, with separate closing and opening thresholds to provide hysteresis. This rule system is understandable and efficient, but universal thresholds may not match one user's hand shape and gesture style.

2 Research goal

The central question was:

Can a lightweight, explainable classical machine-learning model improve static gesture classification for one user across separate recording sessions when compared with the existing hand-written geometric classifier?

The experiment was intentionally framed as a *personalized pilot*. With one participant and one evening available for data collection, it would not be valid to claim population-level or cross-user generalization. The goals were instead to:

- test whether learned boundaries adapt better to the participant;
- compare both classifiers on identical, held-out landmark observations;
- preserve interpretability for later explainable-AI analysis;
- identify gesture classes and failure modes requiring further work; and
- establish a reproducible research workflow without changing the end-user application.

3 Data collection

3.1 Guided collection procedure

A separate playground script guided the participant through randomized gesture trials. For each trial, the interface displayed the target gesture, a countdown, a 1.5-second recording interval, and a rest interval. The nominal protocol used 15 repetitions of each of nine classes, producing 135 prompted trials per session. The participant used the right hand, which is the presentation-control hand in the product.

The nine labels were:

none	point	pinch
pinch_right	open_palm	fist
fist_thumb_up	v_sign	two_finger_scroll

The **none** class represented natural hand motion without an intended command. It is especially important in an interaction system: a classifier that performs well only on carefully posed commands may still trigger actions during ordinary movement.

Three sessions were recorded. Sessions S01 and S02 formed the training data. Session S03 was excluded during training and reserved for the final comparison.

Table 1: Observed dataset size. A trial is a grouped prompted repetition; frames within a trial are temporally correlated.

Session	Frames	Grouped trials	Role
S01	5,284	135	Training
S02	3,509	135	Training
S03	4,762	135	Held-out evaluation
Total	13,555	405	—

3.2 Privacy and stored schema

All camera processing occurred locally. Original images and video were not written to disk. Each stored CSV row represented one detected frame and contained:

- a pseudonymous participant identifier;
- session and trial identifiers;
- the prompted gesture label;

- frame index and relative timestamp;
- handedness and handedness confidence; and
- the (x, y, z) coordinates of all 21 MediaPipe landmarks.

Landmark-only storage substantially reduces privacy exposure and dataset size, although derived hand geometry should not be described as fully anonymous.

4 Feature representation

Each frame was converted into 77 numerical features. First, all landmarks were translated so that the wrist was the origin. Coordinates were then divided by the three-dimensional wrist-to-middle-finger-MCP distance:

$$\tilde{\mathbf{p}}_i = \frac{\mathbf{p}_i - \mathbf{p}_{\text{wrist}}}{\|\mathbf{p}_{\text{middle MCP}} - \mathbf{p}_{\text{wrist}}\|_2}. \quad (1)$$

This produces 63 normalized coordinate features and reduces sensitivity to hand position and apparent size in the camera image.

Fourteen engineered features were appended:

- thumb–index, thumb–middle, and index–middle fingertip distances;
- thumb-tip to index-MCP distance;
- horizontal and vertical thumb displacement;
- normalized index, middle, ring, and little-finger joint angles; and
- the four fingertip distances from the wrist.

These features reflect the geometry already considered by the rule classifier, which makes the learned model suitable for later comparison of manual and learned decision thresholds.

5 Models and the decision-tree rationale

The research tooling supports logistic regression, a random forest, and a decision tree. All three were evaluated, but the decision tree was initially chosen as the primary explainable model for four reasons.

Interpretability. Every prediction follows a path of explicit feature-threshold decisions. It is therefore possible to explain why a frame was classified as a pinch or a fist and to compare learned thresholds directly with the production rules.

Non-linear class boundaries. Gesture classes are defined by combinations of conditions rather than by a single linear separation. A tree can naturally express statements such as a small thumb–index distance together with a sufficiently extended index finger.

Little preprocessing and fast inference. Trees do not require feature standardization and are inexpensive enough for real-time local inference. This is appropriate for a desktop controller.

Appropriate baseline complexity. A random forest may achieve greater robustness but is less transparent because a prediction aggregates many trees. The purpose of this pilot was not only to maximize accuracy, but also to retain a model whose behavior can be inspected.

The fitted tree used a maximum depth of 8, a minimum of 5 training frames per leaf, balanced class weights, and a fixed random seed of 42. Class balancing reduces the influence of unequal numbers of detected frames across gestures. The random forest contained 250 trees, used a minimum of 3 frames per leaf, balanced class weights, and the same seed. Logistic regression used standardized features, balanced class weights, and at most 2,000 optimization iterations. Logistic regression tests whether approximately linear class boundaries are sufficient, while the forest tests the benefit of an ensemble of non-linear trees at the cost of direct interpretability.

6 Experimental design

Each learned model was fitted on 8,793 frames from 270 trials in S01 and S02. No frame from S03 was available during fitting. The serialized model records its training sessions, and the comparison program refuses to evaluate a session that overlaps with them unless the safeguard is explicitly overridden.

For the final comparison, all four classifiers received the same 4,762 held-out landmark frames in the same order. The production `GestureClassifier` was reset at every trial so that its pinch hysteresis could not carry state from one prompted repetition into another.

Two evaluation levels were used:

Frame level: every detected frame was scored separately.

Trial level: frame predictions within a prompted trial were combined using majority voting, and the resulting trial prediction was scored.

Trial-level evaluation is primary because neighboring frames are not independent observations and because a stable interpretation over an entire gesture is closer to the intended interaction. Accuracy, per-class precision, recall, F1 score, and confusion matrices were reported. Precision is the fraction of predictions for a class that are correct; recall is the fraction of true examples of that class that are recovered.

This experiment compares static classification only. It does not compare the complete action pipeline, which additionally includes hold thresholds, debouncing, swipe tracking, scrolling trajectories, and action cooldowns.

7 Results

7.1 Overall performance

Table 2: Held-out performance on session S03.

Classifier	Frame accuracy	Trial accuracy
Random forest	98.8%	98.5%
Logistic regression	95.7%	94.8%
Decision tree	91.7%	93.3%
Hard-coded rules	80.0%	79.3%
Forest-rules difference	+18.8 percentage points	+19.2 percentage points

The forest classified 133 of 135 grouped trials correctly, logistic regression approximately 128, the decision tree approximately 126, and the rules approximately 107. Trial voting improved the

tree’s accuracy relative to its frame accuracy, indicating that many tree errors were brief rather than dominant throughout a repetition. The rule classifier did not show the same benefit.

7.2 Per-class performance

Table 3: Decision-tree trial-level results.

Class	Precision	Recall	F1	Support
fist	1.00	0.93	0.97	15
fist_thumb_up	0.92	1.00	0.96	12
none	0.82	1.00	0.90	14
open_palm	1.00	1.00	1.00	19
pinch	1.00	0.63	0.77	19
pinch_right	0.82	1.00	0.90	14
point	0.92	1.00	0.96	11
two_finger_scroll	1.00	0.94	0.97	18
v_sign	0.93	1.00	0.96	13
Macro average	0.94	0.95	0.93	135
Weighted average	0.94	0.93	0.93	135

Table 4: Random-forest trial-level results.

Class	Precision	Recall	F1	Support
fist	1.00	1.00	1.00	15
fist_thumb_up	1.00	1.00	1.00	12
none	1.00	1.00	1.00	14
open_palm	1.00	1.00	1.00	19
pinch	1.00	0.95	0.97	19
pinch_right	0.93	1.00	0.97	14
point	0.92	1.00	0.96	11
two_finger_scroll	1.00	0.94	0.97	18
v_sign	1.00	1.00	1.00	13
Macro average	0.98	0.99	0.99	135
Weighted average	0.99	0.99	0.99	135

7.3 Confusion matrices

For compactness, the matrices below use class abbreviations: FI (fist), TU (thumb up), NO (none), OP (open palm), PI (index pinch), PR (middle-finger pinch/right click), PO (point), TS (two-finger scroll), VS (V sign), and TL (thumb left). Rows are true classes and columns are predictions.

8 Interpretation

8.1 Comparison of learned models

The random forest was the strongest classifier by a clear margin. It made only two trial-level errors: one index-pinch trial became point, and one two-finger scroll trial became middle-finger pinch. Ensembling reduced the instability of the single tree, especially for index pinch, whose recall rose from 0.63 to 0.95. The trade-off is explainability: feature importance and per-prediction methods can explain a forest, but its decisions cannot be summarized as one short rule path.

Table 5: Logistic-regression trial-level results.

Class	Precision	Recall	F1	Support
fist	1.00	1.00	1.00	15
fist_thumb_up	0.80	1.00	0.89	12
none	0.93	0.93	0.93	14
open_palm	1.00	1.00	1.00	19
pinch	1.00	0.79	0.88	19
pinch_right	0.88	1.00	0.93	14
point	0.92	1.00	0.96	11
two_finger_scroll	1.00	0.89	0.94	18
v_sign	1.00	1.00	1.00	13
Macro average	0.95	0.96	0.95	135
Weighted average	0.96	0.95	0.95	135

Table 6: Hard-coded-rule trial-level results. The rule classifier also produced one `fist_thumb_left` prediction, although this class was not a target in the collected protocol.

Class	Precision	Recall	F1	Support
fist	0.67	0.80	0.73	15
fist_thumb_up	0.80	1.00	0.89	12
none	0.00	0.00	0.00	14
open_palm	0.73	1.00	0.84	19
pinch	0.88	0.79	0.83	19
pinch_right	0.74	1.00	0.85	14
point	0.75	0.55	0.63	11
two_finger_scroll	1.00	1.00	1.00	18
v_sign	1.00	0.85	0.92	13
Macro average including extra predicted class	0.66	0.70	0.67	135
Weighted average	0.74	0.79	0.76	135

Logistic regression ranked second, showing that the normalized and engineered features already separate many gestures with comparatively simple boundaries. Its remaining errors were concentrated in pinch and two-finger-scroll trials. During fitting, the numerical backend emitted transient matrix-overflow warnings. All input features, fitted coefficients, intercepts, and predictions were verified as finite, and optimization completed in 78 iterations. The score is therefore reported, but a clean-environment reproduction is advisable before using it as a final benchmark.

The single decision tree remains the most directly interpretable learned model. Its lower accuracy is the cost of representing the entire classification space with one hierarchy of threshold decisions. Thus model choice depends on the research priority: the forest is currently best for predictive performance, while the single tree is best for direct rule-level XAI.

8.2 Rejection of non-command movement

The most important difference was the `none` class. The tree and forest recovered all 14 held-out `none` trials, and logistic regression recovered 13. The rule classifier recovered none: seven were interpreted as open palm, three as fist, two as pinch, one as point, and one as thumb left. In a live system these errors represent potential false commands, although the production temporal state machines may suppress some short-lived static misclassifications.

This result suggests that a personalized model can learn not only the intended poses but also the participant’s natural non-command hand configurations. The comparison should therefore not be summarized only as an accuracy increase; improved rejection is the more product-relevant finding.

Table 7: Decision-tree trial confusion matrix.

	FI	TU	NO	OP	PI	PR	PO	TS	VS
FI	14	0	1	0	0	0	0	0	0
TU	0	12	0	0	0	0	0	0	0
NO	0	0	14	0	0	0	0	0	0
OP	0	0	0	19	0	0	0	0	0
PI	0	1	2	0	12	2	1	0	1
PR	0	0	0	0	0	14	0	0	0
PO	0	0	0	0	0	0	11	0	0
TS	0	0	0	0	0	1	0	17	0
VS	0	0	0	0	0	0	0	0	13

Table 8: Random-forest trial confusion matrix.

	FI	TU	NO	OP	PI	PR	PO	TS	VS
FI	15	0	0	0	0	0	0	0	0
TU	0	12	0	0	0	0	0	0	0
NO	0	0	14	0	0	0	0	0	0
OP	0	0	0	19	0	0	0	0	0
PI	0	0	0	0	18	0	1	0	0
PR	0	0	0	0	0	14	0	0	0
PO	0	0	0	0	0	0	11	0	0
TS	0	0	0	0	0	1	0	17	0
VS	0	0	0	0	0	0	0	0	13

8.3 Strengths and weakness of the tree

The tree achieved perfect trial recall for thumb up, none, open palm, middle-finger pinch, point, and V sign. It also achieved perfect precision and recall for open palm. Its principal weakness was index pinch: precision was 1.00, but recall was only 0.63. Seven of 19 true pinch trials were assigned to other classes: two to none, two to middle-finger pinch, and one each to thumb up, point, and V sign. Thus the tree was conservative when emitting an index pinch but did not create false index-pinch predictions.

This failure pattern provides a focused XAI question for further analysis: which feature thresholds cause valid pinch frames to follow paths toward these other leaves, and how do those thresholds compare with the manual normalized pinch threshold?

8.4 Strengths and weakness of the rules

The rules performed perfectly on two-finger scroll and recovered every open palm, thumb-up, and middle-finger-pinch trial. However, high recall sometimes coincided with lower precision. Open palm, for example, absorbed seven non-command trials. Point was another major weakness: only six of 11 point trials were recovered, while five were interpreted as middle-finger pinch.

The rules encode useful domain knowledge and remain valuable as an interpretable baseline. Their weakness in this study is not a lack of logic, but the use of fixed boundaries that were not calibrated to this participant or to the observed distribution of natural movement.

9 Validity and limitations

1. **One participant.** The experiment establishes only within-user, across-session performance. It provides no evidence of generalization to other hand shapes, movement styles, skin appearances, cameras, or mobility characteristics.
2. **Static classification only.** The comparison excludes the full temporal action pipeline. Consequently, it cannot yet show a reduction in actual unintended clicks or slide changes.
3. **Correlated frames.** Thousands of frames do not represent thousands of independent examples. Trial-level results are therefore primary.

Table 9: Logistic-regression trial confusion matrix.

	FI	TU	NO	OP	PI	PR	PO	TS	VS
FI	15	0	0	0	0	0	0	0	0
TU	0	12	0	0	0	0	0	0	0
NO	0	0	13	0	0	1	0	0	0
OP	0	0	0	19	0	0	0	0	0
PI	0	2	0	0	15	1	1	0	0
PR	0	0	0	0	0	14	0	0	0
PO	0	0	0	0	0	0	11	0	0
TS	0	1	1	0	0	0	0	16	0
VS	0	0	0	0	0	0	0	0	13

Table 10: Hard-coded-rule trial confusion matrix.

	FI	TL	TU	NO	OP	PI	PR	PO	TS	VS
FI	12	0	3	0	0	0	0	0	0	0
TU	0	0	12	0	0	0	0	0	0	0
NO	3	1	0	0	7	2	0	1	0	0
OP	0	0	0	0	19	0	0	0	0	0
PI	3	0	0	0	0	15	0	1	0	0
PR	0	0	0	0	0	0	14	0	0	0
PO	0	0	0	0	0	0	5	6	0	0
TS	0	0	0	0	0	0	0	0	18	0
VS	0	0	0	2	0	0	0	0	0	11

4. **Trial-identifier reuse in S03.** The nominal protocol specifies 15 trials per class, but the grouped held-out supports range from 11 to 19 while still totaling 135. This pattern is consistent with restarting a randomized collection under the same session and trial identifiers, causing frames from differently prompted repetitions to share a grouping key. The frame labels remain present, but some trial-majority groups are mixed. This weakens the cleanliness of the trial-level estimate. A final confirmatory session should use a fresh session ID and must not be used to retrain or tune the current model.
5. **No statistical significance test.** The report presents a descriptive paired comparison. Confidence intervals or a paired test should be added only after producing clean, uniquely identified held-out trials.
6. **Limited model search.** Three preconfigured classical models were compared without hyperparameter optimization. Evaluating many settings against the same held-out session would itself leak test information, so S03 must not now be used for model tuning.

10 Conclusion and future work

For this participant, all three learned models generalized from two recording sessions to a held-out third session better than the existing geometric rules. The random forest performed best, improving trial-level accuracy by 19.2 percentage points over the rules. Logistic regression ranked second, followed by the single decision tree. Most notably, the learned models distinguished natural non-command motion from command gestures much more reliably. These results support continuing with a personalized learned classifier and XAI analysis, but they are not sufficient to replace the production rules.

The next recommended steps are:

1. collect one clean confirmatory session under a new identifier;
2. export and visualize the tree’s feature importance and decision paths;
3. inspect all misclassified pinch trials and compare learned thresholds with the manual pinch rules;

4. measure false actions per minute after applying both systems' temporal filtering; and
5. if more participants become available, use participant-grouped or leave-one-person-out evaluation.

A Reproduction commands

Each model was trained while reserving S03; the model name and output file were changed for `tree`, `forest`, and `logistic`:

```
venv/bin/python -m research.train \  
  research/data/landmarks.csv \  
  --model tree \  
  --exclude-session S03 \  
  --output research/models/tree.joblib
```

The one-to-one comparison was then run on the held-out session:

```
venv/bin/python -m research.compare \  
  --model research/models/tree.joblib \  
  --session S03 \  
  research/data/landmarks.csv
```

B Scope of the claim

The appropriate conclusion is: *for one participant, three class-balanced classical models trained on two sessions achieved higher static gesture- classification accuracy than the existing geometric rules on a held-out third session, with the random forest achieving the highest accuracy and the single tree offering the most direct explanation.* It would be inappropriate to claim that the models generalize to arbitrary users or that it already improves the complete live application.